

PHYS 2321
Week4: Gauss's Law/Potential

Day 3 Outline

1) Hwk: Ch. 23 P. 2,3,5,9,12,15,17,21,25,28,29,35,36,43,
48,51. MCQ 1-13 odd (Due next Fri)

Next: Read Ch. 23.1-8

2) Ch. 22 Gauss's Law for E near conductors

a. Example: nested metal spherical shells.

3) Ch. 23 Electric Potential

a. Potential, V , of a point charge

b. Electric Potential energy, U , and work

c. Compare to gravitational PE, U_g

Notes: Next quiz is on Mon on Flux and Gauss's Law.

Return Ch. 21B hwk. Mean=9.54/10.

Ch. 22 PDF is my2321wk4.pdf under "NEW STUFF"

Electric Potential of a Point Charge

- Recall E field of point charge:

$$\vec{E} = \frac{kq}{r^2} \hat{r}$$

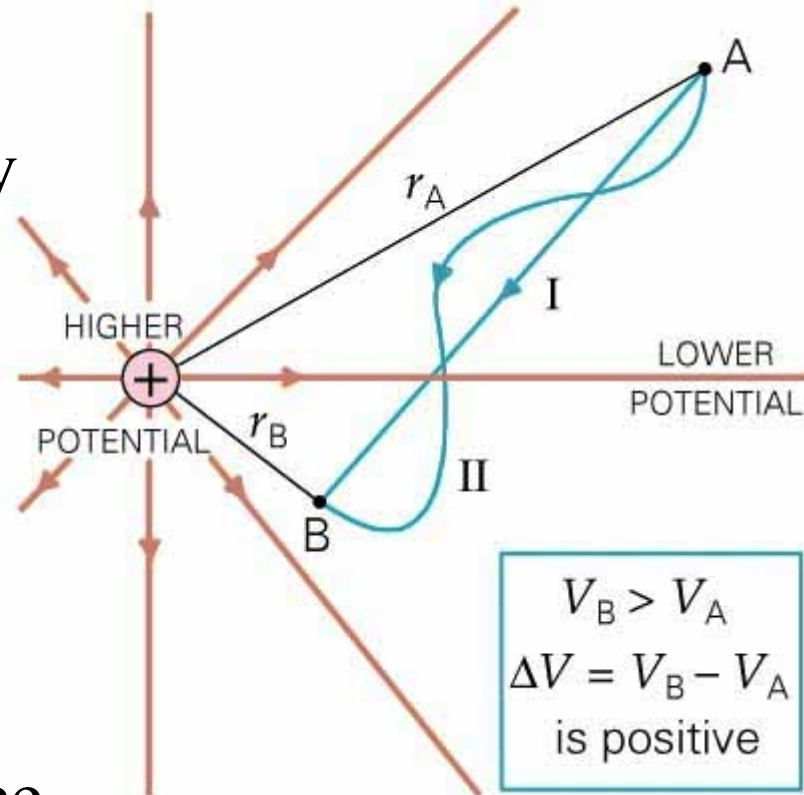
- Electric Potential at r away from a point charge

$$V = \frac{kq}{r}$$

$$V = 0 \quad \text{when} \quad r \rightarrow \infty$$

- Electric potential difference

$$\Delta V_{ab} = \frac{kq}{r_b} - \frac{kq}{r_a}$$



Electric Potential

- Electric Potential of a point charge (previous slide)
- Electric potential closely related to *potential energy*
 - $\Delta U = q\Delta V$
 - And to *work*: $W_{\text{byfield}} = -q\Delta V = -\Delta U$
 - Convention: both U and V = 0 at r=infinity
- Electric potential closely related to electric force
 - $\mathbf{F}_E \cdot \Delta \mathbf{r} = W_{\text{byfield}} = -q\Delta V$
- Electric potential closely related to electric field
 - $\delta V = -E\delta r$ so that potential difference is:
$$\Delta V = - \int \vec{E} \cdot d \vec{l}$$
- Electric potential is easier to work with than the E-field because it is not a vector.

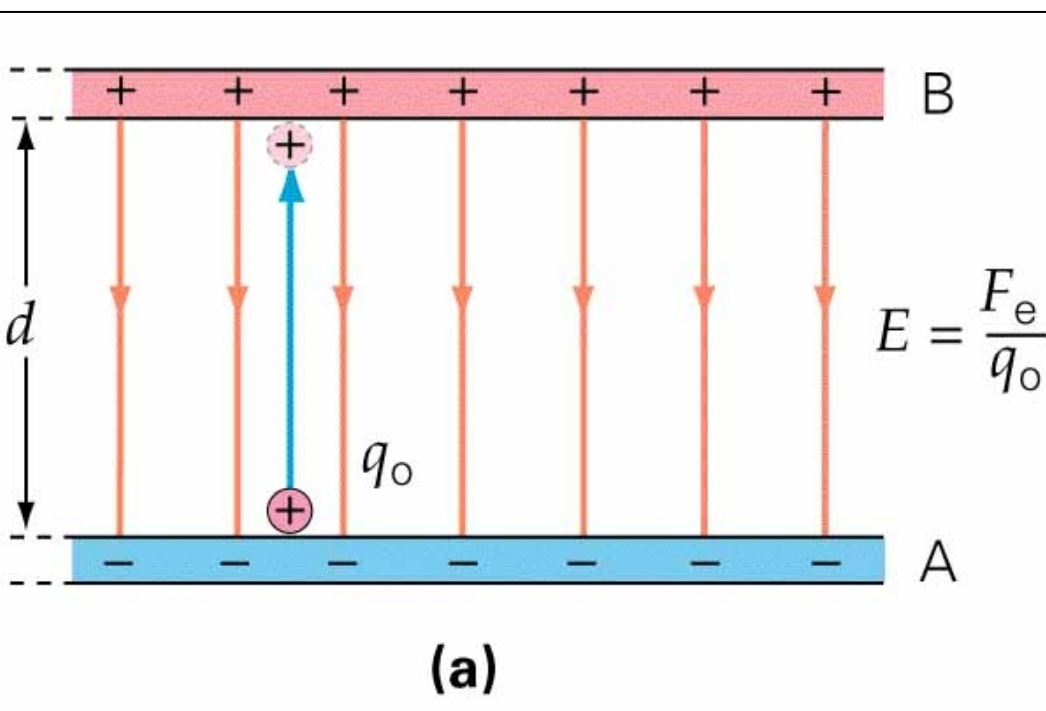
Analogy with gravity

F_E on q_0 is down

$$W_{E\text{field}} = -|F_E|d \quad (F_E = q_0 E)$$

$$\Delta U = -W_{E\text{field}} = |F_E|d$$

$$\Delta V = \Delta U / q_0$$

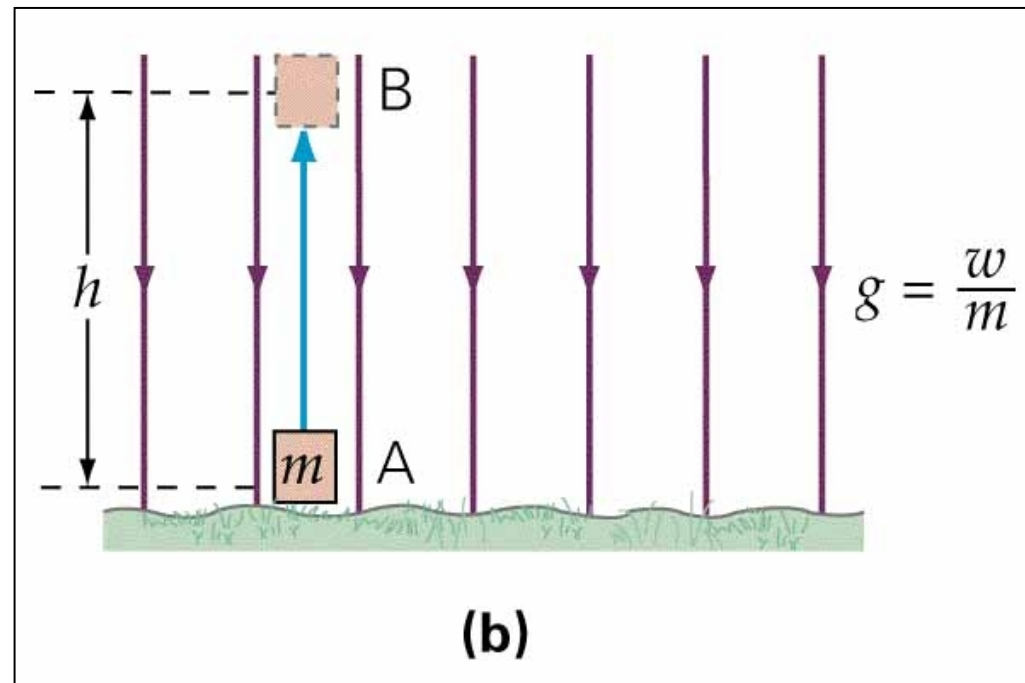


F_g on m is down

$$W_{G\text{field}} = -|F_g|h \quad (F_g = mg)$$

$$\Delta U_G = -W_{G\text{field}} = |F_g|h$$

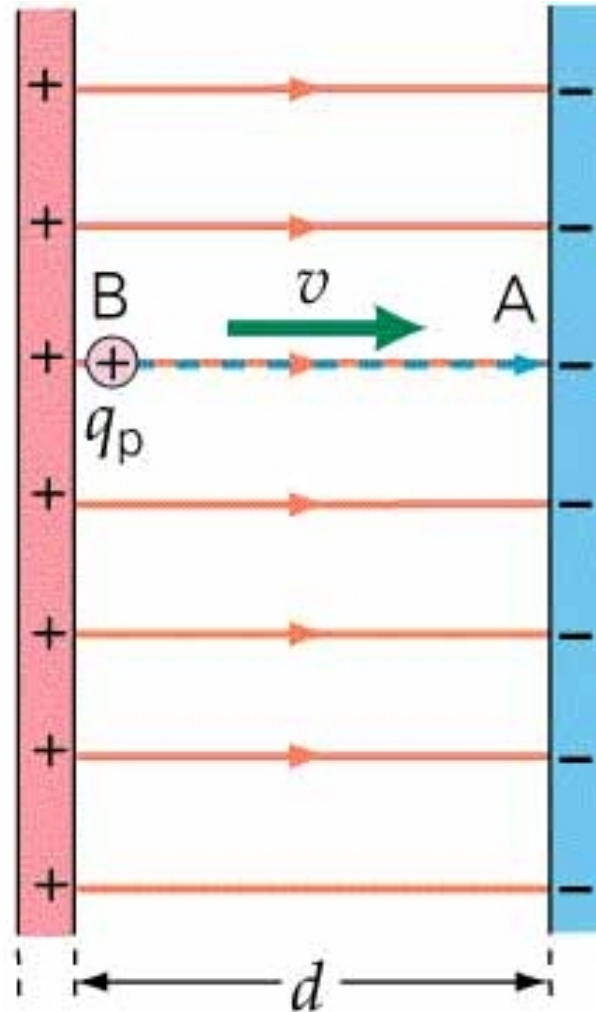
$$\Delta V_G = \Delta U_G / m$$



Parallel Plates

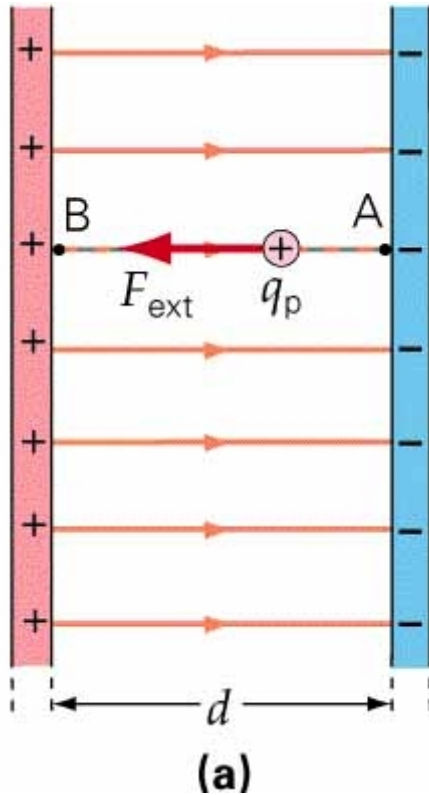
- Releasing a positive test charge from rest at point B...

Find expression
for v at A given
uniform $\mathbf{E}$ and
plate separation d .

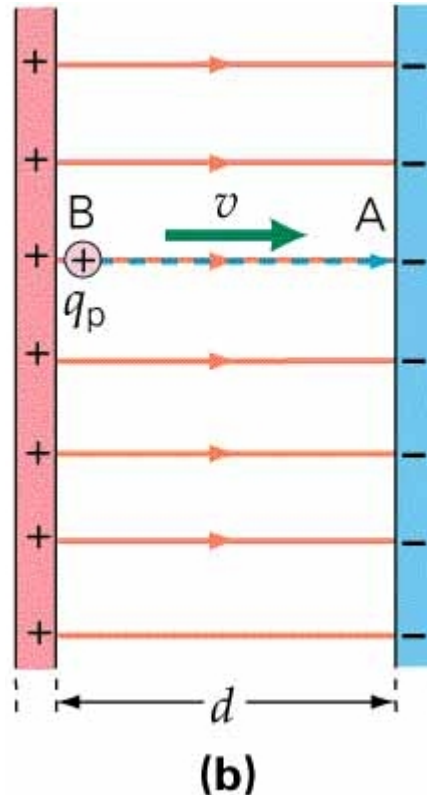


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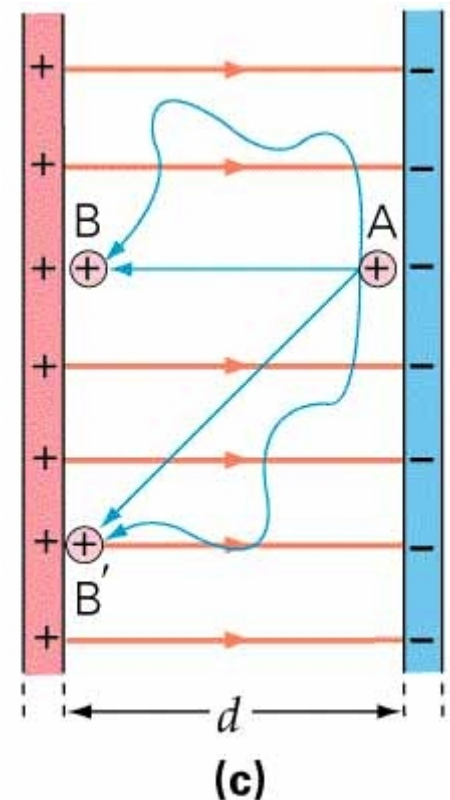
Electric Potential Energy (conservation of energy ideas)



Work is done by “the hand”, so we store potential energy, U_E



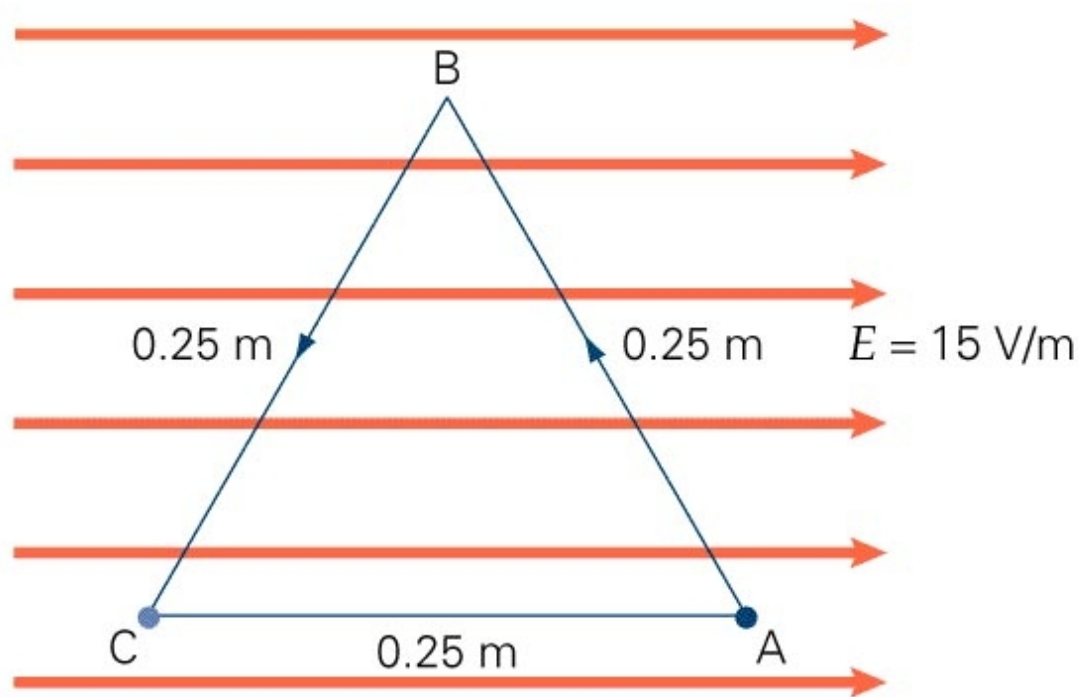
Charge is released and energy is converted from U_E to KE



Only the displacement in the direction of the E field matters
(ΔU_E independent of path)

Problem: closed loop path, ABCA

- Work done is path independent ($\mathbf{F}_E$ is conservative!)
 - Only the initial and final position matter (for $\Delta U, \Delta V, W_{\text{field}}$)
 - Look for an easy solution!
 - True even if E is non-uniform!



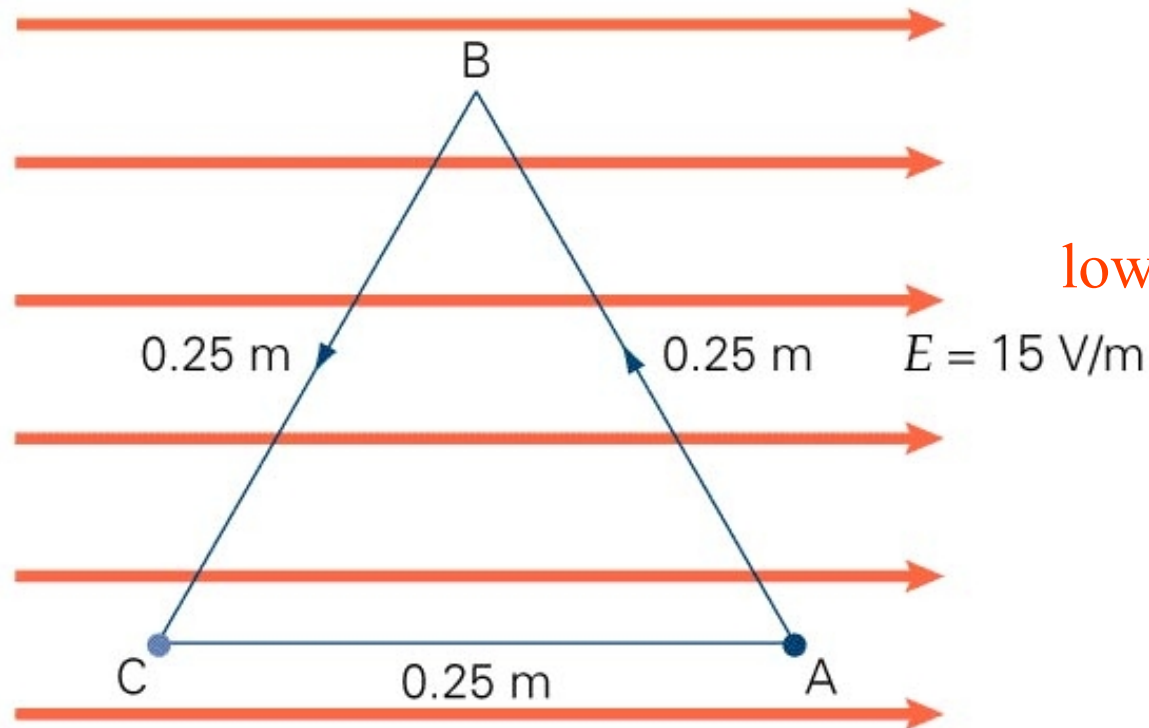
Problem: find ΔV and ΔU along path ABC

Higher V

lower V

Most general:

$$\Delta V = - \int \vec{E} \cdot d\vec{R}$$



$$\Delta V_{ABC} = \Delta V_{AC} = -Ed \cos 180 = \left(15 \frac{\text{V}}{\text{m}}\right)(0.25 \text{ m}) = 3.75 \text{ V}$$

If $q = -e$:

$$\Delta U = q\Delta V = (-1.6 \times 10^{-19} \text{ C})(3.75 \text{ V}) = -6 \times 10^{-19} \text{ J}$$

Problem: find V 's on equipotentials if $V_A = 0$.

$$V_1 - V_5 = 3.75 \text{ V}$$

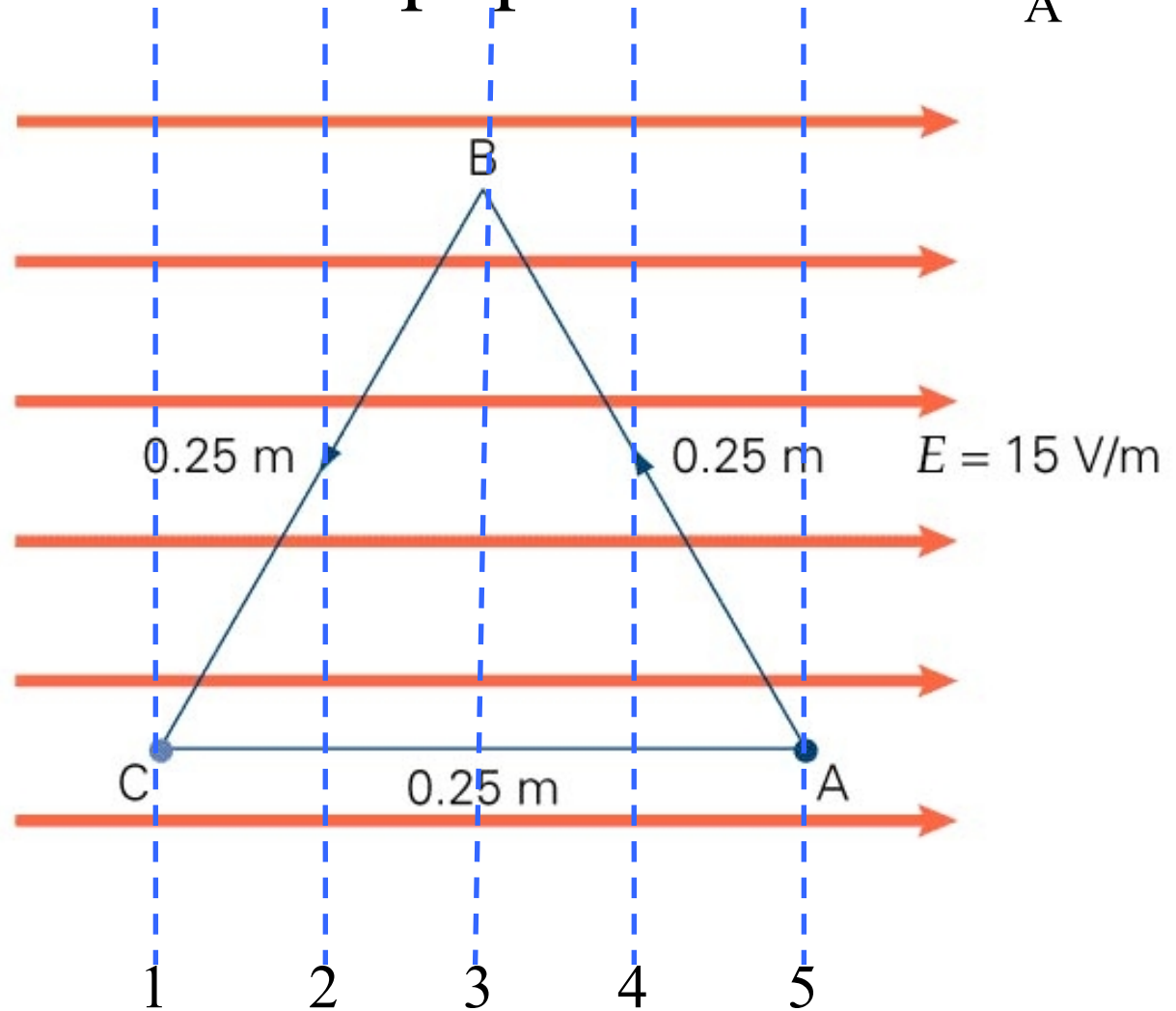
$$V_1 = 3.75 \text{ V}$$

$$V_2 = 2.8125 \text{ V}$$

$$V_3 = 1.875 \text{ V}$$

$$V_4 = 0.9375 \text{ V}$$

$$V_5 = 0 \text{ V}$$



Electric Potential Energy U_E

- Building up arrangements of charge
 - Energy required to “build” = ΔU
- Bring a point charge in from infinity
 - like charges requires energy
 - repulsive forces
 - unlike charges give up energy
 - attractive forces

$$W = Fd = qEd$$

$$\text{and } E = \frac{kq}{r^2}$$

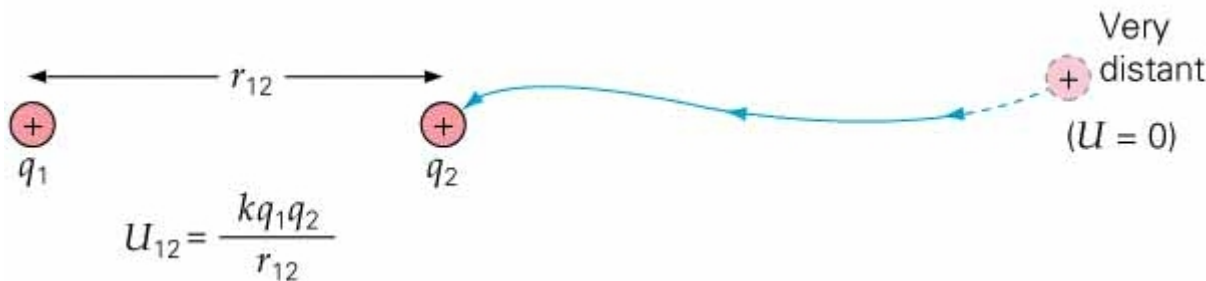
...are difficult to use
since E is not a
constant.

Can use:

$$U_{12} = \Delta U_{12} = q_2 \Delta V_{\infty 1}$$

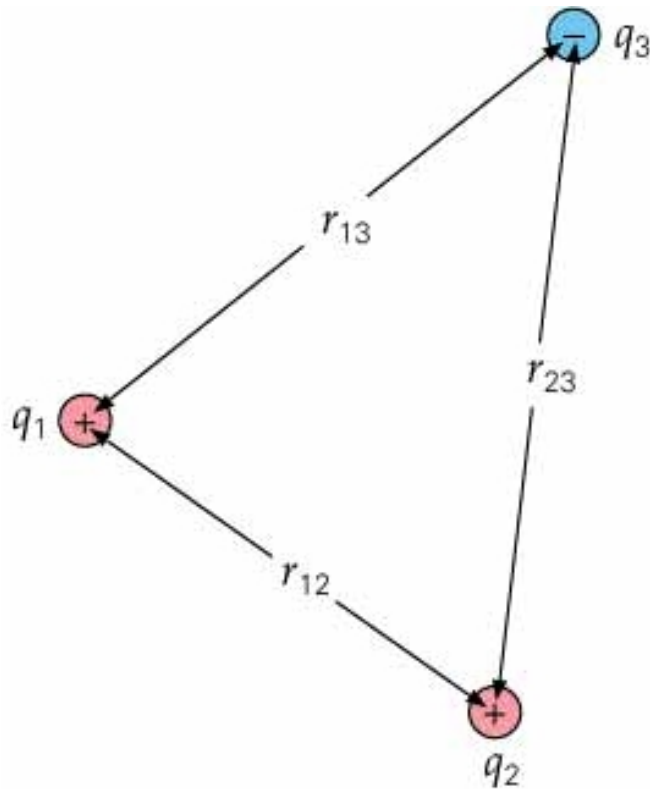
$$V_{\infty} = 0$$

$$V_1 = k \frac{q_1}{r_{12}}$$



U_E for more than two charges

- Don't double count
- Bring each one in from "infinity"
- Bringing together like charges requires energy (force them together)
- Bringing together un-like charges gives up energy (fall together naturally)



$$U_{\text{total}} = U_{12} + U_{23} + U_{13}$$

(b)

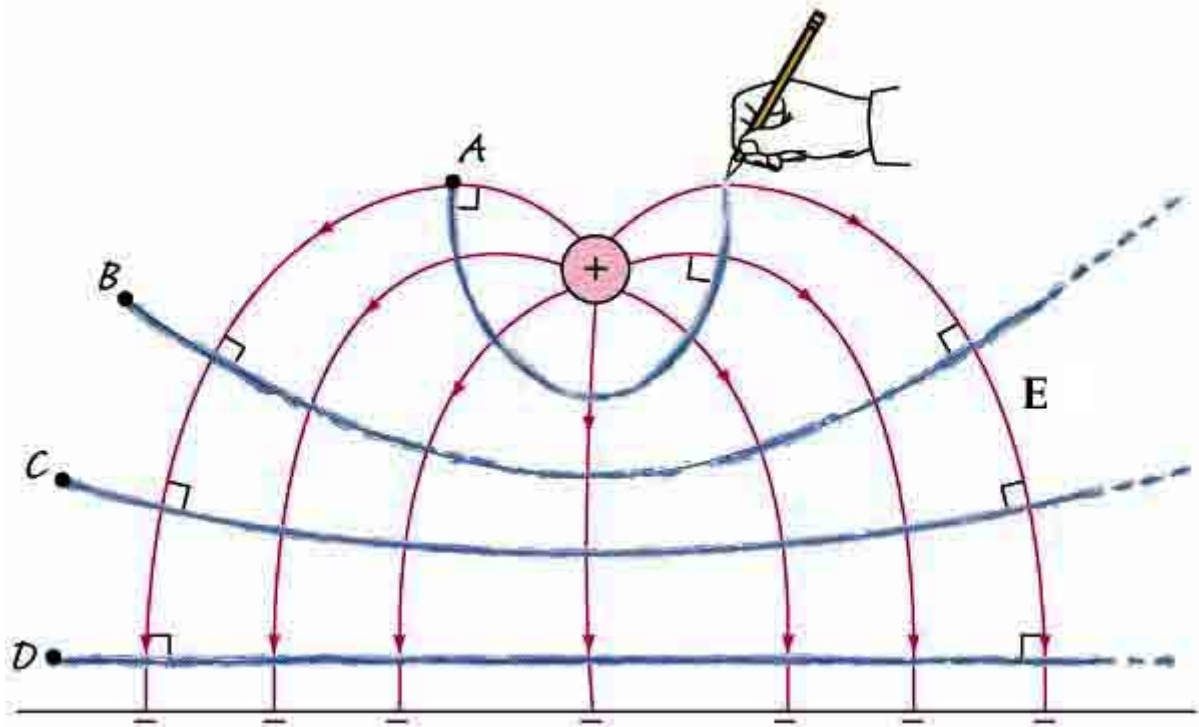
$$U_{12} = k \frac{q_1 q_2}{r_{12}}$$

$$U_{23} = k \frac{q_2 q_3}{r_{23}}$$

$$U_{13} = k \frac{q_1 q_3}{r_{13}}$$

Equipotential Surfaces

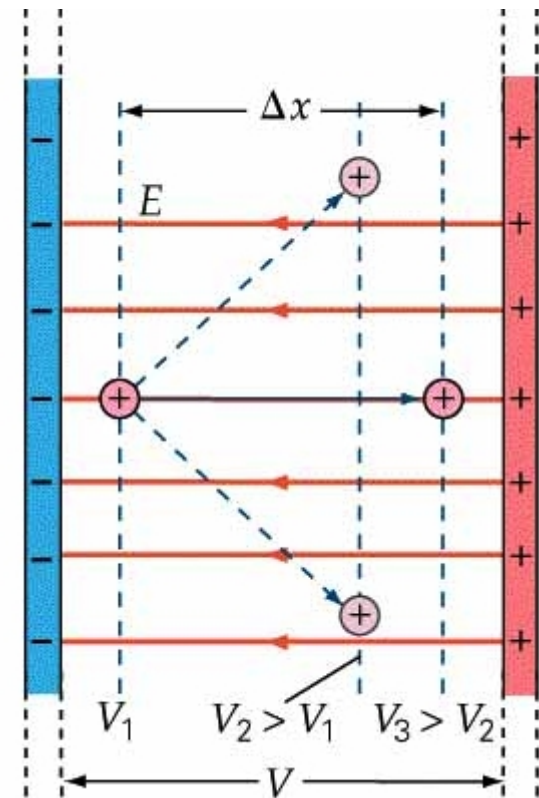
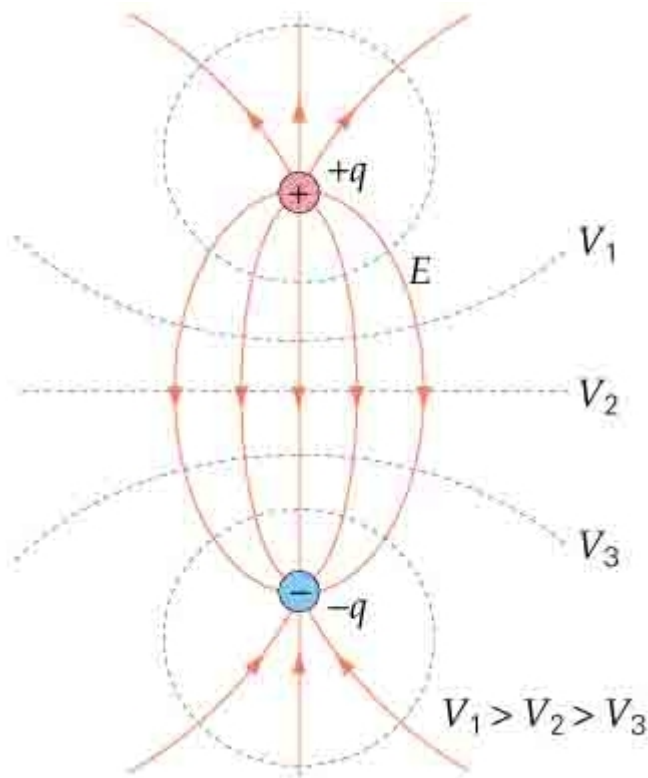
- E field is perpendicular to the equipotential surfaces
- The surface of a conductor is an equipotential surface
 - no E field parallel to the surface in *Electrostatics*
 - gradually “match” the boundaries



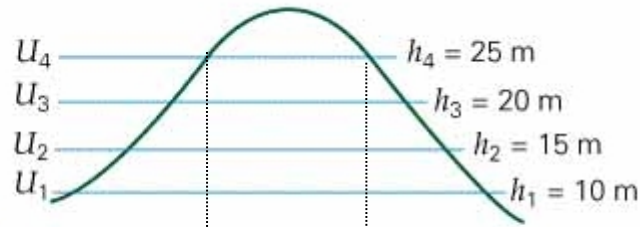
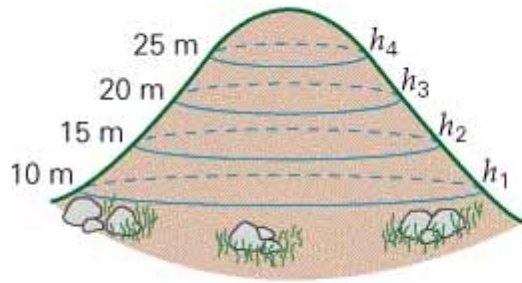
Equipotential Surfaces

Equipotentials are perpendicular to the E-field lines.

E field points “down hill”



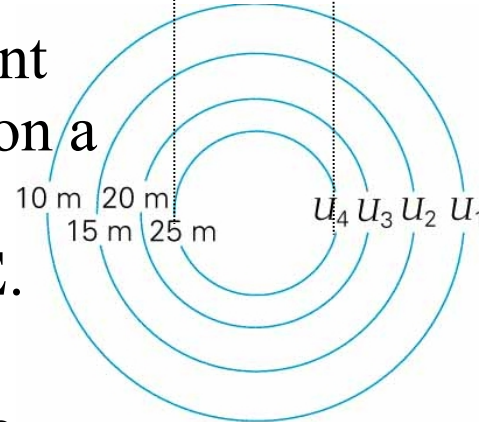
Contours of a map analogy



(a)

Lines of equal altitude are lines of equal gravitational potential energy. This is *like* lines of equal potential.

The net force on a boulder will point downhill. Similarly, the net force on a positive test charge will point “down hill” in direction of $\mathbf{F}$ and $\mathbf{E}$.

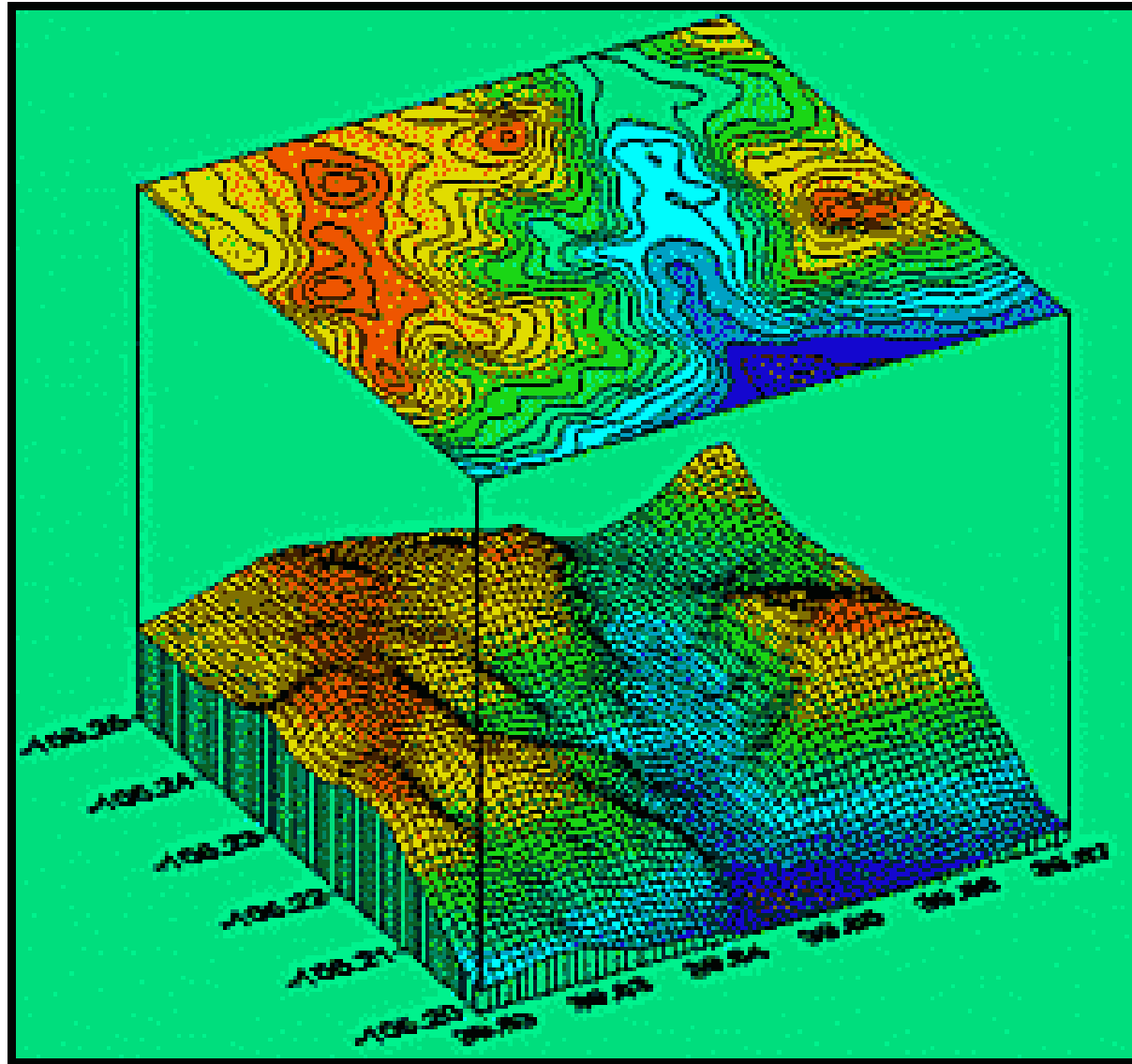


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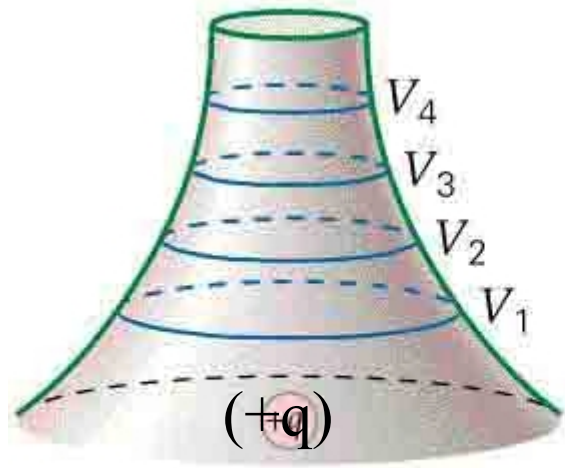
The steeper the hill the stronger the forces and fields!

Expect $\Delta V / \Delta r \sim E$ and F

A topographical map of land.

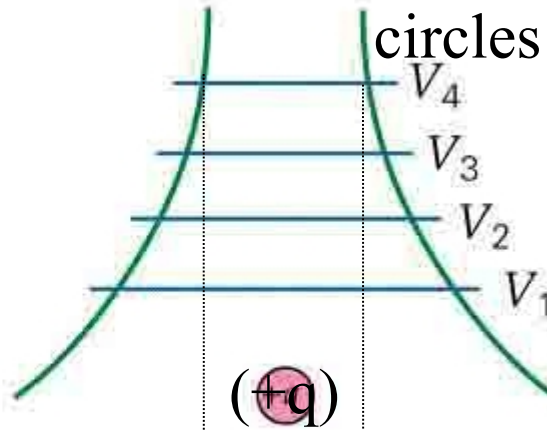


Analogy with Gravity and hills



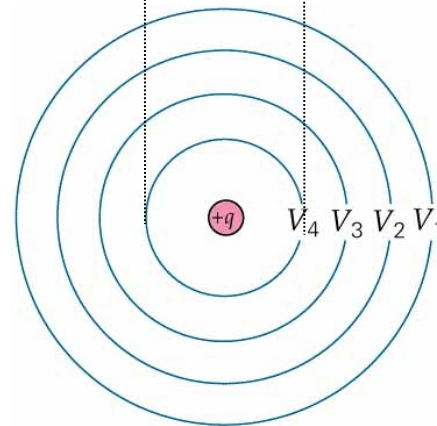
(c)

E field points “down hill”
perpendicular to the
circles



Field gets stronger closer to the
point charge. Don't have to go
as far to have the same change
in electric potential

$$E_r = - \left(\frac{dV}{dr} \right)$$

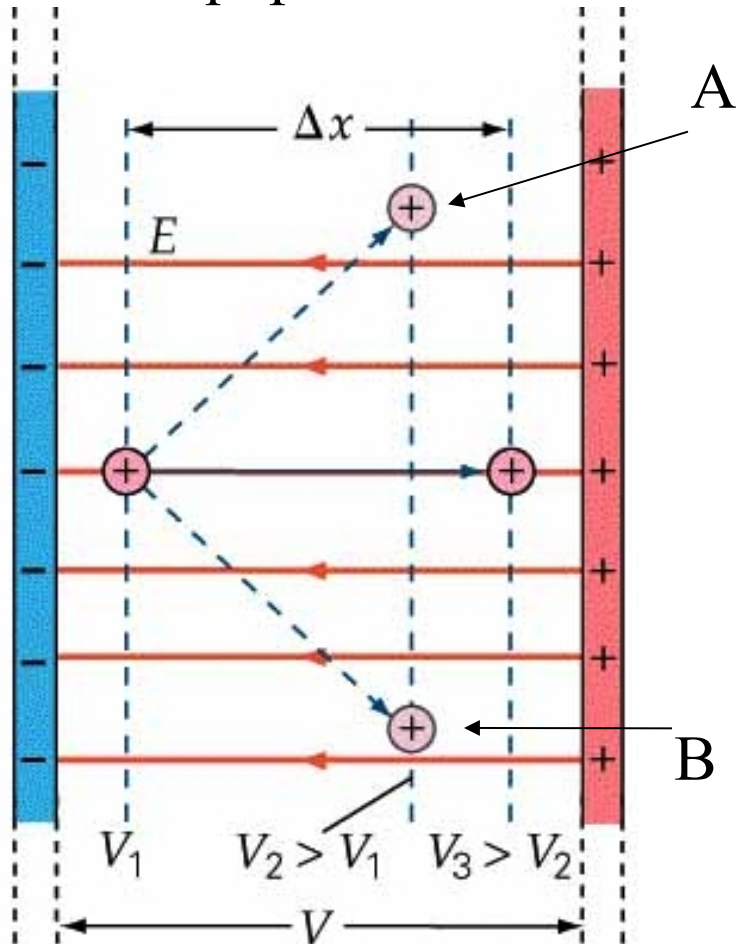


(d)

Slightly
misleading (circles
would not be evenly
spaced for $V \sim 1/r$)

Equipotential Surfaces

- Imaginary or real surfaces of constant voltage
 - The surfaces of a conductor are equipotential surfaces
- E field and equipotential surfaces are perpendicular to each other



If a charge moves from A to B along an equipotential surface, then

$$\Delta V_{AB} = 0$$

$$\Delta U_{AB} = q\Delta V_{AB} = 0$$